

3 Rainbow k -connectivity

In this section, we survey the results on rainbow k -connectivity. By its definition, we know that it is difficult to derive exact value or a nice upper bound of the rainbow k -connectivity for a general graph. Chartrand, Johns, McKeon and Zhang [13] did some basic research on the rainbow k -connectivity of two special graph classes. They first studied the rainbow k -connectivity of the complete graph K_n for various pairs k, n of integers, and derived the following result:

Theorem 3.1 [13] *For every integer $k \geq 2$, there exists an integer $f(k)$ such that if $n \geq f(k)$, then $rc_k(K_n) = 2$.*

They obtained an upper bound $(k+1)^2$ for $f(k)$, namely $f(k) \leq (k+1)^2$. Li and Sun [44] continued their investigation, and the following result is derived:

Theorem 3.2 [44] *For every integer $k \geq 2$, there exists an integer $f(k) = ck^{\frac{3}{2}} + C(k)$ where c is a constant and $C(k) = o(k^{\frac{3}{2}})$ such that if $n \geq f(k)$, then $rc_k(K_n) = 2$.*

From Theorem 3.2, we can obtain an upper bound $ck^{\frac{3}{2}} + C(k)$ for $f(k)$, where c is a constant and $C(k) = o(k^{\frac{3}{2}})$, that is, they improved the upper bound of $f(k)$ from $O(k^2)$ to $O(k^{\frac{3}{2}})$, a considerable improvement. Dellamonica, Magnant and Martin [21] got the best possible upper bound $2k$, which is linear in k (see Theorem 3.3). However, the proof of Theorem 3.2 is more structural or constructive, and informative. In the argument of [21], Dellamonica, Magnant and Martin put forward a new concept, the *rainbow (k, l) -connectivity*.

Given an edge-colored simple graph G , let $l \leq k$ be integers. Suppose the edges of G are k -colored. For $a, b \in V(G)$, denote by $p(a, b)$ the maximum number of internally disjoint rainbow paths of length l having endpoints a and b . The *rainbow (k, l) -connectivity* of G is the minimum $p(a, b)$ among all distinct $a, b \in V(G)$. Note that this new defined *rainbow (k, l) -connectivity* computes the number of internally disjoint paths with the *same* length l (this is distinct from the rainbow k -connectivity which, as mentioned above, computes the number of colors); and by definition, for different edge-colorings, the values of *rainbow (k, l) -connectivity* could be different.

From Theorem 3.1, we know that for any r , there exists an explicit 2-coloring of K_r in which the number of bi-chromatic paths of length 2 between any pair of vertices is at least $\lfloor \sqrt{r} - 1 \rfloor$. Using the above definition, it is a statement about the rainbow (2,2)-connectivity of a given 2-coloring of the edges of K_r . In [21], Dellamonica, Magnant and Martin greatly improve and generalize the above lower bound for graphs of sufficiently large order by providing a different constructive coloring. Their construction attains asymptotically the maximum rainbow connectivity possible.

Theorem 3.3 [21] *For any $k \geq 2$ and $r \geq r_0 = r_0(k)$ there exists an explicit k -coloring of the edges of K_r having rainbow $(k, 2)$ -connectivity*

$$(\frac{k-1}{k} - o(1))r.$$

More generally, They considered the problem of finding longer rainbow paths.

Theorem 3.4 [21] *For any $3 \leq l \leq k$, there exists $r_0 = r_0(k)$ such that for every $r \geq r_0$, there is an explicit k -coloring of the edges of K_r having rainbow $(k, 2)$ -connectivity*

$$(1 - o(1))\frac{r}{l-1}.$$

This result is also asymptotically best possible, since any collection of internally disjoint paths of length l can contain at most $\frac{r}{l-1}$ paths. Their proof employed a very recent breakthrough due to Bourgain [8, 51], which consists of a powerful explicit extractor. Roughly speaking, an (explicit) extractor is a polynomial time algorithm used to convert some special probability distributions into uniform distributions. See [54] for a good but somewhat outdated survey on extractors.)

Chartrand, Johns, McKeon and Zhang [13] also investigated the rainbow k -connectivity of r -regular complete bipartite graphs for some pairs k, r of integers with $2 \leq k \leq r$, and they showed:

Theorem 3.5 [13] *For every integer $k \geq 2$, there exists an integer r such that $rc_k(K_{r,r}) = 3$.*

However, they could not show a similar result for complete graphs, and therefore they left an open question: For every integer $k \geq 2$, determine an integer (function) $g(k)$, for which $rc_k(K_{r,r}) = 3$ for every integer $r \geq g(k)$, that is, the rainbow k -connectivity of the complete bipartite graph $K_{r,r}$ is essentially 3. In [45], Li and Sun solved this question using a similar but more complicated method to that of Theorem 3.5, and they proved:

Theorem 3.6 [45] *For every integer $k \geq 2$, there exists an integer $g(k) = 2k \lceil \frac{k}{2} \rceil$ such that $rc_k(K_{r,r}) = 3$ for any $r \geq g(k)$.*

More generally, we have the following question.

Question 3.7 [13] *Does the following hold? For each integer $k \geq 2$, there exists an integer $h(k)$ such that for every two integers s and t with $h(k) \leq s \leq t$, we have $rc_k(K_{s,t}) = 3$.*

Recently, He and Liang [29] investigated the rainbow k -connectivity in the setting of random graphs. They determined a sharp threshold function for the property $rc_k(G(n, p)) \leq d$ for every fixed integer $d \geq 2$. This substantially generalizes a result due to Caro, Lev, Roditty, Tuza and Yuster (see Theorem 2.46). Their main result is as follows.

Theorem 3.8 [29] *Let $d \geq 2$ be a fixed integer and $k = k(n) \leq O(\log n)$. Then $p = \frac{(\log n)^{1/d}}{n^{(d-1)/d}}$ is a sharp threshold function for the property $rc_k(G(n, p)) \leq d$.*

The key ingredient of their proof is the following result: With probability at least $1 - n^{-\Omega(1)}$, every two different vertices of $G(n, \frac{C(\log n)^{1/d}}{n^{(d-1)/d}})$ are connected by at least $2^{10d} c_0 \log n$ internally disjoint paths of length exactly d .

So far, results on the rainbow k -connectivity are just those for two special graph classes, and a sharp threshold function for the property $rc_k(G(n, p)) \leq d$ where $d \geq 2$ is a fixed integer and $k = k(n) \leq O(\log n)$. Clearly, there are a lot of things one can do further on this concept. As mentioned above, it is difficult to derive the precise value or a nice upper bound for $rc_k(G)$ of a κ -connected graph G , where $2 \leq k \leq \kappa$. So one may consider the following problem.

Problem 3.9 *Derive a sharp upper bound for $rc_2(G)$, where G is a κ -connected graph with $\kappa \geq 2$. Does $rc_2(G) \leq \alpha n$, where $0 < \alpha < 1$ is independent of n ?*

4 k -rainbow index

The k -rainbow coloring as defined above, is another generalization of the rainbow coloring. In [15], Chartrand, Okamoto and Zhang did some basic research on this topic. There is a rather simple upper bound for $rx_k(G)$ in terms of the order of G , regardless the value of k .

Proposition 4.1 [15] *Let G be a nontrivial connected graph of order $n \geq 3$. For each integer k with $3 \leq k \leq n - 1$, $rx_k(G) \leq n - 1$ while $rx_n(G) = n - 1$.*

There is a class of graphs of order $n \geq 3$ whose k -rainbow index attains the upper bound of Proposition 4.1, it is an immediate consequence of Observation 2.11.

Proposition 4.2 [15] *Let T be a tree of order $n \geq 3$. For each integer k with $3 \leq k \leq n$, $rx_k(T) = n - 1$.*

There is also a rather obvious lower bound for the k -rainbow index of a connected graph G of order n , where $3 \leq k \leq n$. The *Steiner distance* $d(S)$ of a set S of vertices in G is the minimum size of a tree in G containing S . Such a tree is called a *Steiner S -tree* or simply a *Steiner tree*. The k -*Steiner diameter*, say $sdiam_k(G)$ of G is the maximum Steiner distance of S among all sets S with k vertices in G . Thus if $k = 2$ and $S = \{u, v\}$, then $d(S) = d(u, v)$ and the 2-Steiner diameter $sdiam_2(G) = diam(G)$. The k -Steiner diameter then provides a lower bound for the k -rainbow index of G : for every connected graph G of order $n \geq 3$ and each integer k with $3 \leq k \leq n$, $rx_k(G) \geq sdiam_k(G) \geq k - 1$.